Sociotechnical Security

CS 8803/4803

Professor Veronica Rivera

Lecture 13 – Scams II (Impact of AI, solution spaces)



Content warning: This lecture and discussion mention emotional harm in romantic/close relationships & human trafficking

Last time

- Started discussing scams
  - 4 primary areas of focus in digital safety on scam research
- Heeyun presented on two papers
  - Scambaiting using honeypots on Craigslist
  - Interviews with victims of pig-butchering scams to identify the scam lifecycle
- Discussion
  - Different interventions at different stages
  - Interventions for trust-building scams need to be cross-platform
  - Interventions should consider what social need leads people to fall for scams
    - E.g., loneliness
  - How should we better engage with each other surrounding scams
    - E.g., telling friends about digital interactions; creating ways for people with limited social networks to do so too

Today

- Approaches used to study scams
- Role of AI in scams
- Solution spaces
- Open problems
- Raj leading discussion

Approaches used to study scams

Research focus	Methods and evidence
Why people engage	Interviews, surveys, and lab/field studies of scam judgments Social media and forum analysis
How scams operate	Victim interviews Infrastructure measurement & analysis of insider accounts Direct scammer engagement and analysis of messages and conversations (scam-baiting)
Victim experiences and harms	Victim interviews Social media and forum analysis Surveys of financial, emotional, and social impacts
Protection and victim support	Interviews with victims and support network (e.g., friends, family) User studies and field tests of warnings, education, and interventions Co-design with at-risk groups Evaluation of detection, reporting, and support services

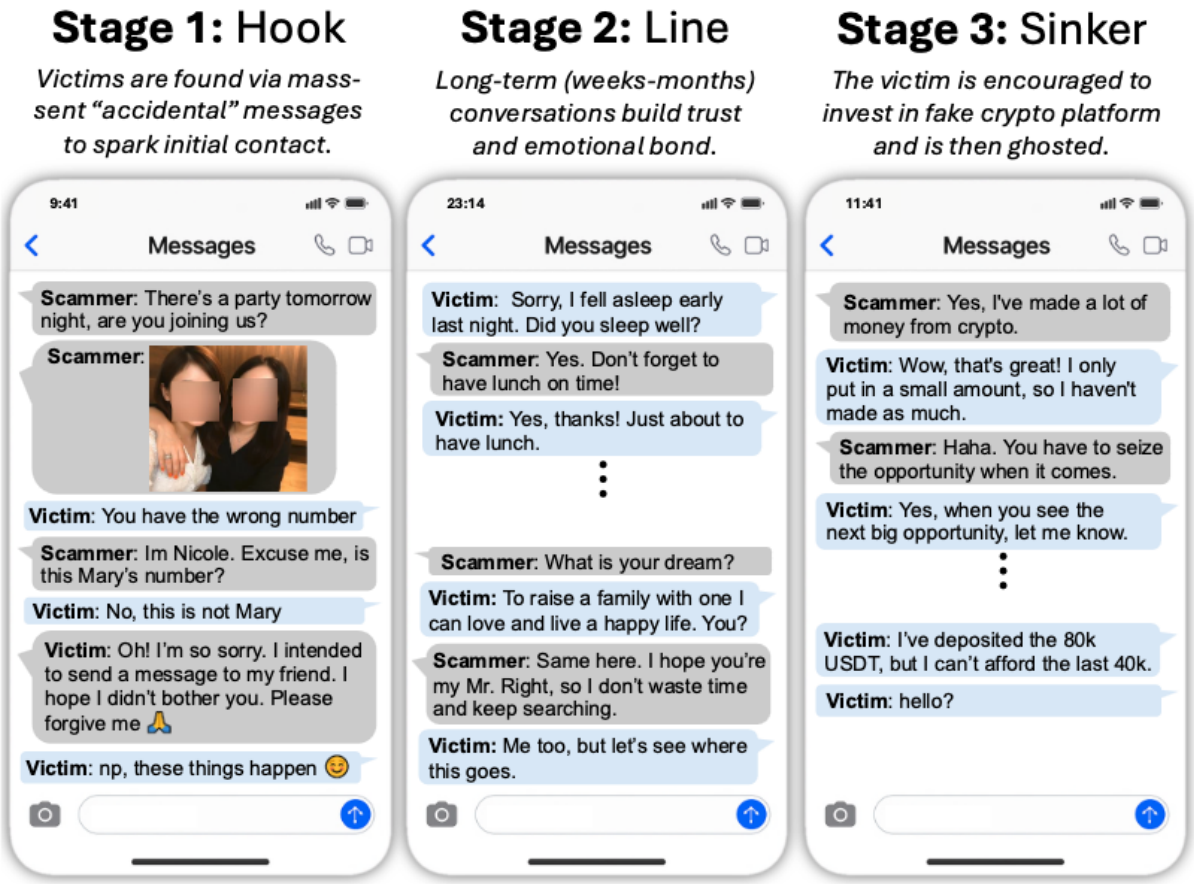
Recall: AI in scams (phishing)

- LLMs can generate phishing content cheaply and at scale [Hazell, 2023]
  - LLMs can produce realistic phishing emails and websites for cheap while evading model safeguards
- AI-generated emails get the same click-through rate as emails written by human experts [Heiding et al., 2024]
  - Click-through rates for human-generated (54%) and LLM-generated (56%) were similar
  - Using LLMs can increase profitability up to 50%
- (Broader) LLMs are persuasive and can pass as human
  - Some models have been shown to be more persuasive than a paid human [Liu et al., 2026]
  - Even more persuasive when they have information about you [Salvi et al., 2025]
  - --> increased sophistication of attacks

Role of AI in romance-baiting scams?

How much of romance-baiting (e.g., pig-butchering) scams are and will be automated using LLMs?

RQ1: Feasibility of full automation
RQ2: Adversarial capacity
RQ3: Quantifying the threat
RQ4: Trust harvesting
RQ5: Safe guards



Gressel et al., Love, Lies, and Language Models: Investigating AI's Role in Romance-Baiting Scams, USENIX Security 2026

Approach & results: RQ1 and RQ2

How are LLMs used to automate these attacks and how much of the current infrastructure can be replaced with LLMs

Approach

- Interviews (from 2022-2025)
  - 115 low-level scammers
  - 30 high-level personnel
  - 5 victims
- Focused on scam compounds in Southeast Asia

Results

- 87% of people in the scam compound do labor that AI could easily take over.
- From 2024 onwards, all insiders reported using ChatGPT in scams
  - Tone/fluency polishing, translation to other languages, drafting replies
- Final extraction is still done by humans

Approach & results: RQ3 and RQ4

Can LLMs surpass human operators at carrying out romance-baiting scams?

Approach

- Deception study
  - 22 participants from a university
  - 7 day interaction with human and LLM agent partner
  - On day 7, each partner was told to convince the participant to install a benign mobile app
  - Do they do it?

Results

- 20/22 participants correctly identified the LLM partner, only in hindsight.
- LLM partners persuaded more participants to download the benign app than humans
  - 46% for LLMs vs. 18% for humans

Approach & results: RQ5

Do current LLM safeguards prevent this type of misuse?

Approach

- Safeguard auditing
  - Will models admit they are AI when asked?
    - (OpenAI's GPT-4o, Google's Gemini 2.5 Pro, and Anthropic's Claude 3.7)
  - Do content filters catch scams?
    - (Meta's Llama Guard 3, OpenAI's Moderation API, and Google's Perspective API.)

Results

- All three tested models denied being AI when asked
- Commercial content filters failed to flag romance-baiting dialogues
  - Reason: Scam messages looked like friendly, supportive chat

How victims perceive AI used to scam them

Approach: Qualitative content analysis, timeline analysis, and descriptive statistics of scam victimization across 7 subreddits

- r/Scams, r/scammers, r/Sextortion, r/phishing, r/Cryptoscams, r/catfish, r/Romancescam

Early results:

- Three uses of AI in scams
  - AI used as a rhetorical device
  - AI used to justify behavior
  - AI used to automate scams
    - Advertising
    - Fake profiles
    - AI-generated receipts
    - AI in product
    - AI in customer service
- Victims come up with folk theories of how AI works to explain their experiences
- Victims also use AI to deal with the fallout
  - Using AI to gather stories of others' experiences and find commonalities

Work in progress led by Jasmin Wyss (RUB)

Solution spaces

Scam detection/mitigation

- Education and training
  - Scams beyond phishing & email
  - For vulnerable groups (e.g., older adults)
  - Training using GAI (Hoffman et al., 2026)
- Warnings
  - Context-specific explanations in SMS using GAI (Bouma-Sims et al., 2026)

Victim-survivor support

- Social and community support
  - What support do victims seek?
- Connecting victim-survivors with resources
  - How do victim-survivors recover and what challenges do they experience? (Chen et al., 2026)
  - → (Open) What role could GAI play in victim-survivor support?

Apple Impersonation Risk Detection

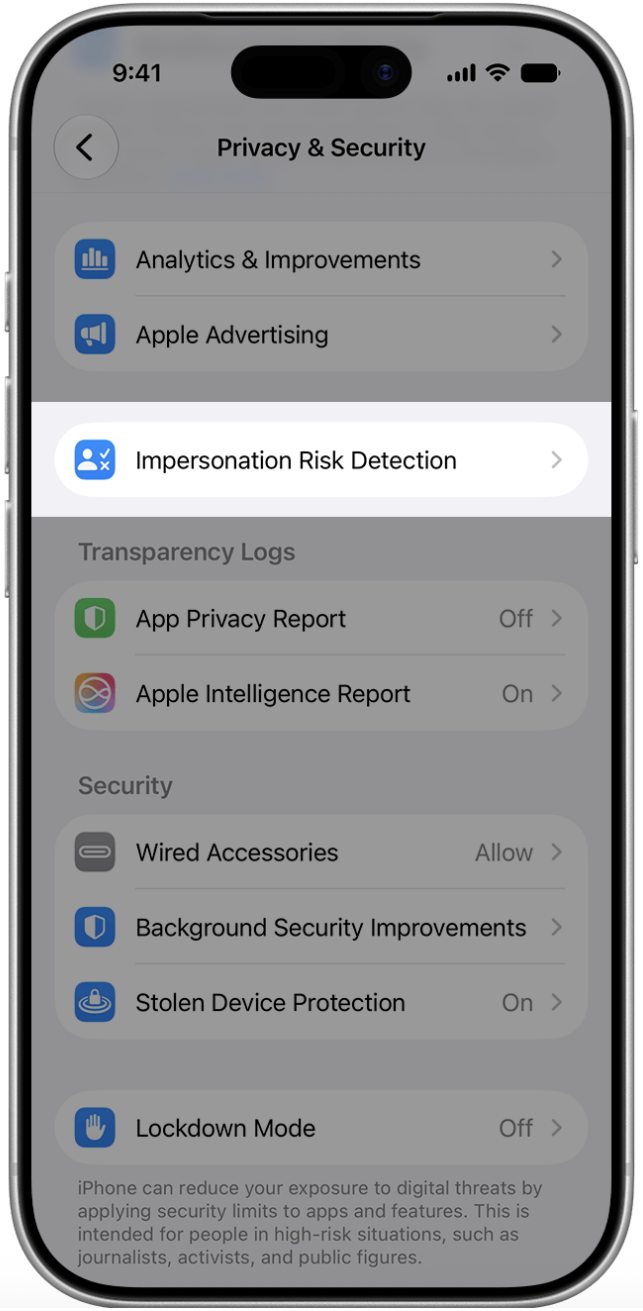
Impersonation scams (e.g., social engineering) get past traditional security (e.g., 2FA) because the victim performs the action themselves.

- Defense needs to detect signs of manipulation

In iOS 27 and iPadOS 27

https://support.apple.com/en-us/127906

- Analyzes information about your device and Apple account
- Returns a risk level to the app
- App receives the risk level, not the information used to generate it
- App decides what to do next
  - Verify identity, add a delay, show a warning



Some open problems in scams research

- Which types of scams are most susceptible to LLM automation?
- Are the cues for LLM-supported impersonation the same as for human-supported impersonation?
- Detection of scams that rely on trust over long periods of time
- What role should generative AI play in supporting scam victim-survivors?
  - Triaging across different stakeholders in the support process?
  - What are the privacy implications of such a tool, and how should they be communicated to users?
- Responsibility across platforms and organizations for detection and support

Extra credit opportunity: Attend SCP Colloquium on Oct 22 11am

https://sites.gatech.edu/talksatscp/?page_id=8

Talk title: Crime Exponentiation is a Wicked Problem: Why Cybersecurity and Criminology Need each Other to Solve Agentic Crime

Speaker: Volkan Topali (https://ays.gsu.edu/profile/volkan-topalli/)

Professor of Criminal Justice and Criminology at GSU



- Talk is on Oct 22 at 11am in the Coda 9th floor Atrium
- Free lunch at the talk!
- Registration link will be posted at the website soon (and on Ed)
- **If you attend and say hi to Veronica after the talk, you'll get 5 points added to the reading reflection due that week**
- Email me if you would like a 30 minute meeting with him

No class next Monday 10/19

- Veronica will be away at an NSF workshop
- Midterm project report due at 2pm